

EMMANUEL KATIPUNAN

katipunan.emmanueljr@gmail.com | +63 964-849-9165 | <https://emmsdev-portfolio.vercel.app/>

PROFESSIONAL SUMMARY

Motivated and impact-driven Software Developer with 3+ years of professional experience building scalable backends, full-stack web applications, and system integrations.

Proficient in Object-Oriented Programming (OOP) principles, C#, .NET Core frameworks, Python, and SQL databases. Skilled in software debugging, RESTful API development, and unit testing, with a strong background in delivering maintainable and reliable enterprise software solutions.

TECHNICAL SKILLS

- **Programming Languages:** C#, PHP, JavaScript, TypeScript, Python, SQL
- **Backend & APIs:** ASP.NET Core, .NET, PHP/Laravel, Node.js, Express.js, RESTful API Development, API Integration
- **Frontend:** React.js, Vue.js, HTML5, CSS3
- **AI & LLM Engineering:** LLM Integration, Retrieval-Augmented Generation (RAG), AI Agent Orchestration, Semantic Search, Prompt Engineering, AI-powered Applications
- **Databases:** MySQL, PostgreSQL, MongoDB, Redis, Firebase
- **Cloud & Deployment:** Amazon Web Services (AWS), Google Cloud Platform (GCP), Cloud Application Deployment, Server Deployment, CI/CD, Docker
- **Software Engineering:** Object-Oriented Programming (OOP), SOLID Principles, Clean Architecture, Microservices, Software Debugging & Troubleshooting, System Integration, Unit Testing
- **Desktop Development:** C#/.NET Desktop Applications
- **Development Practices & Tools:** Git, Agile/Scrum

WORK EXPERIENCE

Software Developer

Callbox Inc. | *June 2023 – August 2026*

- Designed, developed, and maintained core software integrations and serverless application services following strict OOP and clean architecture principles.
- Engineered high-performance RESTful APIs and real-time data sync processes to maintain system stability and eliminate data loss under heavy load.
- Collaborated with cross-functional development teams in an Agile environment, actively engaging in code reviews, sprint execution, and technical documentation.
- Troubleshoot and debugged complex software issues, implementing automated processing tools that optimized internal workflow efficiency by 50%.

PROJECTS & ACADEMIC HIGHLIGHTS

AI-Powered Knowledge Base System (C#, ASP.NET Core, RAG)

- Developed a knowledge base system using **C# and ASP.NET Core** for centralized management, retrieval, and querying of organizational knowledge.
- Integrated **AI-powered semantic search and Retrieval-Augmented Generation (RAG)** to retrieve relevant information from stored documents and provide context-aware responses.
- Designed RESTful APIs and backend services for document processing, knowledge retrieval, authentication, and AI-assisted querying.
- Implemented a RAG pipeline combining document retrieval, contextual data injection, and AI-generated responses to improve response relevance and accuracy.

Smart File Explorer (C#, ASP.NET Core, RAG, LLM)

- Developed an AI-powered file management and discovery system using **C# and ASP.NET Core**, allowing users to create, organize, and manage files and folders.
- Integrated **Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG)** to enable natural-language file search and conversational interaction with stored documents.
- Implemented document retrieval and contextual processing so the AI can identify relevant files and generate responses based on the user's stored knowledge.

- Designed natural-language search capabilities that allow users to locate files based on their **content and context**, rather than relying solely on filenames or folder locations.

Real-Time Application Architecture (YourLife & EchatMo)

- Developed real-time database-backed applications with secure authentication, API interfaces, and optimized cloud data storage solutions.

Applicant Hiring Portal — Capstone Project

- Developed an online Applicant Hiring Portal as an academic capstone project to streamline the recruitment and hiring process for the university.
- Built a job posting portal where applicants can browse available positions, submit online applications, and upload required documents.
- Implemented applicant profile and application management features that enable hiring panels to efficiently review candidate information, applications, and submitted requirements.
- Integrated an online interview workflow, allowing hiring panels to conduct applicant interviews directly through the platform.
- Designed the system to centralize recruitment activities and reduce manual processing throughout the application and candidate evaluation lifecycle.

EDUCATION

Bachelor of Science in Information and Communication Technology

West Visayas State University - Calinog Campus | *June 2019 – March 2023*

- **Honors & Awards:** Best Programmer Awardee | Best Thesis Awardee
- **Key Coursework:** Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems (SQL), Software Engineering Practices

CHARACTER REFERENCES

- **Gill Cunanan** — *Frequentis*
Email: gillx.cunanan@gmail.com | Phone: 09165623111
- **Lucky John Faderon** — *Outsource Accelerator*

Email: lfaderon@gmail.com | Phone: 09499967617

- **Louie Marte Doromal** — *Callbox Inc.*

Email: ldbopz17@gmail.com | Phone: 09381309230